SECTION A: MULTIPLE CHOICE QUESTIONS - CHOOSE ONLY ONE CORRECT AND BEST ANSWER FOR EACH QUESTION AND CIRCLE THE LETTER IN FRONT OF IT. ANSWER ALL QUESTIONS (40 x 1 = 40 MARKS)

1. ....is an example of covalent modification used mainly for regulation of enzyme activity
   a. Amination
   b. Acetylation
   c. Phosphorylation
   d. Methylation
   e. Decarboxylation

2. Which of the following statements is correct about most enzymes?
   a. They increase the activation energy so that the reaction occurs
   b. They can make a reaction happen faster
   c. There are used once and need to be replaced after catalyzing a reaction
   d. They are stable in both high acid and high alkaline solutions
   e. They are not specific to the type of substrate they receive

3. Which one of the following statements concerning glutamine is correct?
   a. Contains an amide group
   b. Has Glu as its three letter abbreviation
   c. Classified as an acidic amino acid
   d. Has a charged polar side chain
   e. Doesn't have a chiral carbon

4. Succinate dehydrogenase catalyses the dehydrogenation of succinate to fumarate. Malonic acid is used to inhibit the action of this enzyme. What type of inhibition is this?
   a. Competitive
   b. Uncompetitive
   c. Non competitive
   d. Feed back
   e. Allosteric

5. ....is associated with a condition known as Alkaptonuria.
   a. Proline
   b. Methionine
   c. Phenylalanine
   d. Histidine
   e. Valine

6. An amino acid that yields acetoacetyl CoA during catabolism of its carbon skeleton is...........

a. Glucogenic
b. Ketogenic
c. Hydrophobic
d. Polar
e. Polar and hydrophobic

7. ....is an amino acid the contains an imidazole ring

a. Tyrosine
b. Phenylalanine
c. Histidine
d. Proline
e. Tryptophan

8. A 50-year-old patient is brought to the emergency department with the diagnosis of myocardial infarction. Which lactate dehydrogenase (LDH) isoenzyme activity would prevail in the patient's serum in the first two days after hospitalisation?

a. LDH_1
b. LDH_2
c. LDH_3
d. LDH_4
e. LDH_5

9. A chiral carbon of an amino acid arises from the fact that it.....................................

a. is symmetric
b. is bonded to both amino and carboxylic groups
c. is bonded to four different chemical groups
d. has net charge of zero
e. assumes an L configuration

10. The stability of protein structures is supported by a number of bonds. Which bonds support the primary structure?

a. Ionic
b. Peptide
c. Hydrogen
d. Hydrophilic
e. Hydrophobic

11. The normal hemoglobin of an adult person consists of two identical α and two identical β polypeptide chains. What is this level of structure called?

a. Primary
b. Secondary
c. Quaternary
d. Tertiary
e. Binary

12. One of the common laboratory experiments in Biochemistry determines electrophoretic spectrum of plasma proteins. What is the physiochemical properties of proteins utilized in this method?
   a. Inability to denaturation
   b. Presence of charge
   c. Optical activity
   d. Solubility
   e. Viscosity

13. Amino acids are an important substrate for gluconeogenesis during fasting. Which one of the following is a major gluconeogenic amino acid received by the liver?
   a. Alanine
   b. Glycine
   c. Cysteine
   d. Pyruvate
   e. Tryptophan

14. Which amino acid is involved in producing glutathione, an important antioxidant?
   a. Glutamine
   b. Glutamate
   c. Proline
   d. Ornithine
   e. Arginine

15. A protein's activity is altered when a particular serine side chain is phosphorylated. Which of the following amino acid substitutions at this position could lead to a permanent alteration in normal enzyme activity?
   a. Serine to Glutamic acid
   b. Serine to Threonine
   c. Serine to Tyrosine
   d. Serine to Lysine
   e. Serine to Leucine

16. One of the main sources of non-volatile acid in the body is sulfuric acid generated from the sulfur-containing compounds in ingested food or from the metabolism of the sulfur-containing amino acids. Which of the following amino acids would lead to sulfuric acid formation?
   a. Cysteine and isoleucine
   b. Cysteine and alanine
   c. Cysteine and methionine
   d. Methionine and isoleucine
   e. Isoleucine and alanine

17. What would be the most likely composition of amino acids that form a portion of a transmembrane protein located within the plasma membrane?
   a. acidic
   b. basic
   c. hydrophobic
   d. glycosylated
   e. acetylated

18. Five isoenzymic forms of lactate dehydrogenase from the human blood serum were isolated and their properties studied. What property indicates that the isoenzymic forms were isolated from the same tissue?
   a. Same physiochemical properties
   b. Catalyzation of the same reaction
   c. The same electrophoretic mobility
   d. Tissue localisation
   e. Same molecular weight

19. Which one of the following can influence the charge of amino acid residues in the active site of an enzyme?
   a. Competitive inhibitor
   b. Excess substrate
   c. pH of medium
   d. Temperature
   e. Atmospheric pressure

20. Regulation of enzyme activity is the basis for feedback inhibition in enzyme catalysis. Which one of the following represents the most common type of regulation?
   a. Covalent modification
   b. Allosteric regulation
   c. Proteolysis
   d. Carboxylation
   e. Hydroxylation

21. Which of these is an example of the secondary structure of a protein?
   a. Peptide bonding between amino acids
   b. Hydrophobic interactions
   c. Hydrogen bonding between R groups
   d. Disulphide bonds between cysteine residues
   e. Hydrogen bonding an amine and carbonyl group

22. Which level of protein structure is determined only by hydrogen bonds?
   a. Primary
   b. Tertiary
   c. Secondary
   d. Quaternary
   e. Ovoidal

23. Which globin chains are affected in glycosylated Hemoglobin?
   a. γ-globin chains
   b. α-globin chains
   c. β-globin chains
   d. δ-globin chains
   e. both γ and α-globin chains

24. Patients, homozygous for hemoglobin C, have a condition that arises from.........................
   a. substitution of lysine by glutamic acid in the β-chain
   b. substitution of glutamic acid by lysine in the β-chain
   c. substitution of valine by glutamic acid in the β- chain
   d. substitution of a polar amino acid by a non-polar one in the β-chain
   e. substitution of one of the α-chains by a β-chain

25. Which one of these statements is correct about Thalassemias?
   a. hereditary hemolytic diseases in which there is an imbalance in the synthesis of globin chains
   b. they are polygenic disorders
   c. only α-globin chains are affected
   d. they are not in a class of hemoglobinopathies
   e. Only γ- chains are affected

26. Which two amino acids in collagen usually undergo post translational modification?
   a. lysine and histidine
   b. glycine and alanine
   c. valine and isoleucine
   d. lysine and proline
   e. leucine and glutamic acid

27. Lack of post translational modifications of some amino acids in collagen leads to non-cross linkage of the fibers. Which one of the following refers to the condition that arises?
   a. sickle cell anemia
   b. pernicious anemia
   c. scurvy
   d. phenylketonuria
   e. hemosiderosis

28. Which of the following is responsible for specifying the 3-dimensional shape of a protein?
   a. The peptide bond
   b. The amino acid sequence
   c. Interaction with other polypeptides
   d. Interaction with molecular chaperons
   e. Interaction with water molecules

29. Which one of the following plasma proteins is the most abundant in blood?
   a. Globulins
   b. Albumin
   c. Fibrinogen
   d. Prothrombin
   e. Thrombin

30. What is the primary function of albumin in plasma?
   a. Blood clotting
   b. Transport of lipids
   c. Maintaining osmotic pressure
   d. Antibody production
   e. Transport carbohydrates

31. Which plasma protein is involved in blood clotting?
   a. Albumin
   b. Globulins
   c. Fibrinogen
   d. Complement
   e. C-reactive protein

32. Which plasma protein is produced primarily by the liver?
   a. Immunoglobulins
   b. Fibrinogen
   c. Albumin
   d. Complement
   e. Thrombin

33. What are globulins primarily responsible for?
   a. Transporting nutrients
   b. Blood clotting
   c. Immune response
   d. Maintaining blood volume
   e. Maintaining oncotic pressure

34. Sickle cell anemia is caused by a mutation in the gene that codes for:
   a. Myoglobin.
   b. Hemoglobin.
   c. Red blood cells.
   d. Albumin
   e. Globulins

SECTION B: DESCRIPTIVE ANSWER TYPE/SCENARIO BASED QUESTIONS. ANSWER ALL QUESTIONS. FOR EACH ANSWER, ILLUSTRATIONS WITH DIAGRAMS (WHERE POSSIBLE) WILL ATTRACT MAXIMUM MARKS (40 MARKS)

1. Plasma proteins are composed mainly of glycoproteins, lipoproteins and immunoglobulins. These plasma proteins are involved in binding and transporting nutrients, toxic substances etc. These proteins can be divided broadly into three types with each type performing a specific function

    i. What is the normal reference range (in grams/dl) for total plasma proteins? (1 mark) 1

    60 — 70 g/L or 6.0 — 7.0 g/dL

    ii. What is the term given to the soluble fraction of blood that lacks coagulation factors? (1 mark)

    Serum

    iii. Besides clotting factors, which other protein lacks in the soluble fraction referred to, in (ii)? (1/2 mark)

    Albumin, globulin

    iv. How are normal or reference ranges for biochemical parameters established? (2 marks)

    - Collect blood samples from a healthy population
    - then they statistically analyze the blood
    - they then establish the mean and standard deviation of the blood samples
    - They then establish the reference range.

    1½

    5½

    v. Which one is a major plasma protein and what percentage of total plasma protein does it account for? (1/2 mark)

    ½ Albumin accounts for 55% — 60%

    vi. Why is the protein in (v) assessed in patients suspected to have liver disorders? (1/2 mark)

    - The levels of it in blood showcases or confirms the disorder
    - Abnormal levels of it indicates a disorder.

    vii. Name at least one function of protein in (v) and explain how it carries out this function (1/2 mark)

    - Providing capillary osmotic pressure
    - It attracts other ions ~~hence~~ through osmosis hence balancing the levels and causing osmotic pressure.

    ½

viii. Which plasma protein is elevated during inflammation and infections? (1 mark)

1 C-reactive protein (CRP)

ix. What is its classification? (1 mark)

Acute phase protein

1

x. Where are most plasma proteins synthesized? (1 mark)

Liver

1

xi. Name four properties of proteins that are used in separating plasma proteins (2 marks)

- Net charge
- Size
- Shape
- Solubility

2

2. A new drug is developed which selectively cleaves covalent bonds between two sulfur atoms of non-adjacent amino acids in a polypeptide chain.

i. Which level of protein structure in affected molecules would be most directly affected by the drug? (1 mark)

~~tertiary level~~

ii. Explain why this level is affected (1 mark)

~~- Because it has disulphide bonds as one of the cardinal or important forces stabilizing it.~~
~~- Disulphide bonds stabilize tertiary structure~~

iii. Describe, with the help of diagrams, the other levels of protein structure (3 marks)

[redacted]

[redacted]

iv. Which forces or interactions maintain levels of protein structures? Note: For each level, indicate the force/interaction (2 marks)

Primary level: Peptide bonds
Secondary level: hydrogen bonds
Tertiary level: Disulphide bonds,
Quaternary level: Van der wals or London dispersion forces

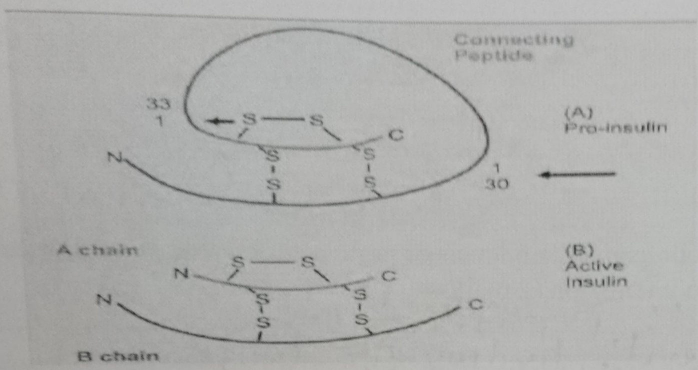
v. Which information is responsible for specifying the three dimensional structure of a protein? (1 mark)

Amino acid sequence or primary structure information.

vi. Name the amino acid that is capable of forming disulphide bonds in between peptide chains (1 mark)

Cysteine

3. Beta cells of the pancreas synthesize insulin as a prohormone. Proinsulin is a single polypeptide chain with 86 amino acids. Biologically active insulin (2 chains) is formed by removal of the central portion of the pro-insulin before release.



For questions below, refer to the figure above

i. Why does the pancreas produce insulin as a prohormone? (1 mark)

[redacted]

ii. How is the prohormone converted to an active hormone? (1 mark)

[redacted]

iii. What happens to the connecting polypeptide indicated in the figure, in the process of activation? (1 mark)

[redacted]

iv. What is the general term for enzymes that cut at the sites indicated by arrows? (1 mark)

[redacted]

v. Classify (using the main amino acid classification), the amino acid responsible for producing the S-S linkages in the figure (1 mark)

[redacted]

vi. Explain the fact that species variation in insulin is restricted to position 8, 9, 10 in the A chain and to position 30 in the B chain (2 marks)

[redacted]

vii. Sequence homology is the term used to show that human insulin is similar to porcine insulin. Can porcine insulin then be used on diabetic human beings. Justify you answer (2 marks)

[redacted]

viii. Write the full name of a peptide formed by carboxyl group of glycine with amino group of alanine and then carboxyl group of alanine with amino group of valine? (1 mark)

4. Rates of enzymatic reactions are influenced by a variety of factors, such as concentration of substrates, products, inhibitors, temperature and pH levels. Study of these parameters is important because it provides a great deal of information on nature of a particular enzyme and the biochemical pathways involved

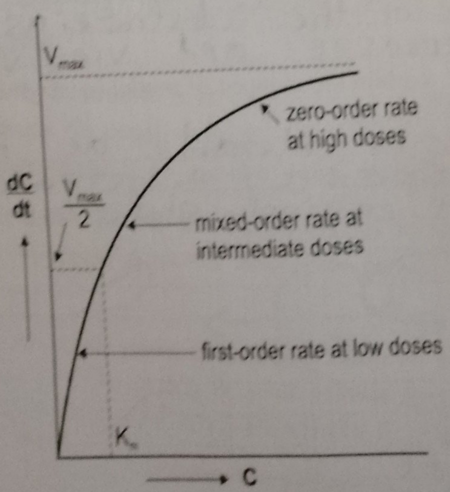


Fig: Plot showing the effect of substrate concentration on velocity of a reaction
Explain portions indicated as:

i. First order (2 marks)

- Substrate concentration is low
- The reaction velocity and substrate concentration is direct proportional at this point

ii. Mixed order (2 marks)

- 50% of enzymes are bound to the substrates
- The reaction velocity is half of the maximum reaction velocity (V.max)

iii. Zero order (2 marks)

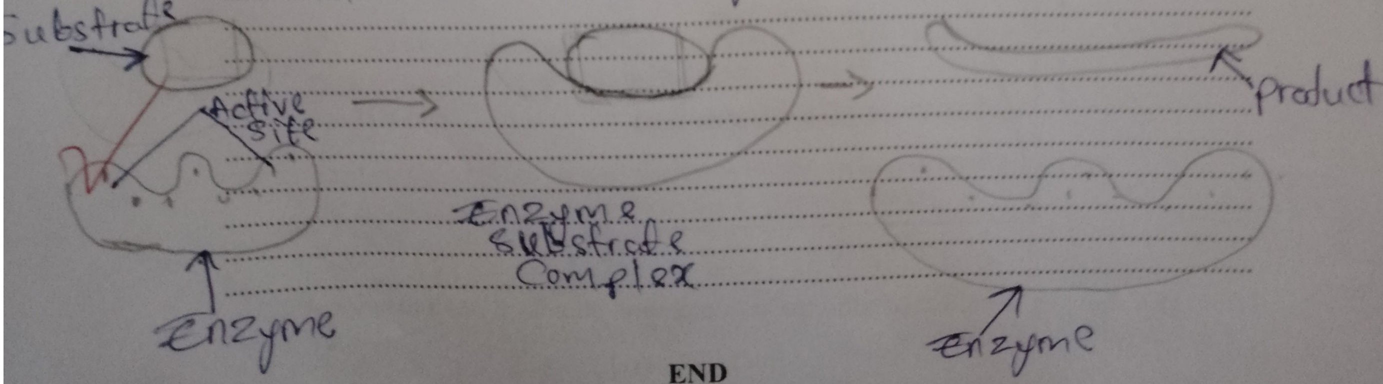
- Shows maximum velocity of reaction (Vmax)
- All the enzymes are bound to the substrates
- The enzymes are saturated as there more substrates hence no increase in reaction velocity

iv. Define Km. How does it relate to the affinity of the enzyme for the substrate? (2 marks)

- Michaelis constant (Km) is the substrate concentration that occurs at half of ~~the reaction velocity~~ maximum reaction velocity
- $Km \propto 1/affinity$: Michaelis constant is ~~direct pro~~ inversely proportional to the enzymes affinity of the substrate, as Km increases enzymes affinity decreases and vice versa.

v. With the help of some diagrams, explain the induced fit model in enzyme catalysis (2 marks)

- The active site is flexible as it changes according to a specific substrate it binds and Returns to its original state after reaction.



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